Q4 Architecture Spec: Payments API v3

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1. Overview

Payments API v3 replaces the v2 gateway for card and UPI payments. It moves partner traffic to the new API gateway, adds the FastPay Gateway processor for UPI, and consolidates logging into a single pipeline. Target launch: 15 December 2026.

2. Authentication and authorization

- All partners authenticate with OAuth 2.0 client credentials issued by the Acme auth service.
- Access tokens expire after 60 minutes. Refresh tokens rotate on every use.
- Internal services use Workload Identity; no static keys in production.

3. Network and transport security

Endpoint	Consumers	Protocol	Notes
api.acme.in/v3/payments	Mobile apps, web	TLS 1.3	Public, behind API gateway
partners.acme.in/v3	Partner banks	TLS 1.2+	Mutual TLS required
legacy-settle.acme.in	Two legacy settlement partners	TLS 1.1	Kept enabled until partners upgrade (no date agreed)
Internal service mesh	Internal services	mTLS (TLS 1.3)	Managed by service mesh

4. API protection

- Per-client token-bucket rate limiting at the gateway: 100 requests per second, burst 200; HTTP 429 with Retry-After.
- Request payloads limited to 1 MB.

5. Data storage and residency

- Payment records in Cloud Spanner, asia-south1, encrypted with CMEK keys rotated every 90 days.
- Daily backups to asia-south2.

6. Logging and retention

Log type	Destination	Retention
Application logs	Cloud Logging (default bucket)	30 days
Security and access audit logs	Cloud Logging (default bucket)	30 days, to reduce storage cost
Payment transaction ledger	Spanner + archival	8 years

7. Third-party vendor: FastPay Gateway Ltd (UPI processing)

Service level	FastPay commitment
Monthly availability	99.5%
Severity 1 response	Within 1 hour (business hours), 4 hours (nights and weekends)
Severity 1 resolution	Within 8 hours
Breach notification	Within 24 hours
Service credits	5% of monthly fee below 99.5%
Data location	Mumbai and Singapore data centres

8. Open items

- Confirm decommission date for legacy-settle.acme.in.
- Finalize FastPay contract (legal review in progress).
- Load test at 1,000 requests per second before launch.